Solution of Triangle: Formula Sheet

Basic laws

Sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Cosine rule:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Cot in sides:

$$\cot A = \frac{b^2 + c^2 - a^2}{4\Delta}$$

Projection: $a = b\cos C + c\cos B$

Tangent rule:

$$\tan\frac{B-C}{2} = \frac{b-c}{b+c}\cot\frac{A}{2}$$

Nature (c largest): $a^2 + b^2 \gtrless c^2$

Area and half angles

Area:

$$\Delta = \tfrac{1}{2}bc\sin A = \frac{abc}{4R} = rs$$

Heron: $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

Altitude:

$$p_a = \frac{2\Delta}{a}$$

$$\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$$

$$\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$$

$$\tan\frac{A}{2} = \frac{\Delta}{s(s-a)} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

Ambiguous case and m-n theorem

Third side:

$$a = c\cos B \pm \sqrt{b^2 - c^2\sin^2 B}$$

Count: $b < c\sin B$: 0; $=$: 1; $c\sin B < b < c$: 2; $b \geq c$: 1

m-n:

$$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$$

m-n: $(m+n)\cot\theta = n\cot B - m\cot C$

Circumcircle and incircle

Circumradius:

$$R = \frac{a}{2\sin A} = \frac{abc}{4\Delta}$$

O to BC: $R\cos A$; $\angle BOC = 2A$

Inradius:

$$r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2}$$

Inradius:

$$r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$IA = r\operatorname{cosec}\frac{A}{2}$; $\angle BIC = \frac{\pi}{2} + \frac{A}{2}$

Tangent lengths: $s-a,\ s-b,\ s-c$

Bisector: $BD : DC = c : b$; $AI : ID = (b+c) : a$

Excircles

Exradius:

$$r_1 = \frac{\Delta}{s-a} = s\tan\frac{A}{2}$$

Exradius:

$$r_1 = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$$

$I_1A = r_1\operatorname{cosec}\frac{A}{2}$, $I_1B = r_1\sec\frac{B}{2}$

$$\angle BI_1C = \frac{\pi}{2} - \frac{A}{2}$$

Sums: $\sum r_1 = 4R + r$, $\sum r_1 r_2 = s^2$

$rr_1r_2r_3 = \Delta^2$; $\sum\frac{1}{r_1} = \frac{1}{r}$

Tangent to excircle from A: s

Median, bisector, orthocentre

Median: $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$; $AG : GD = 2 : 1$

Angle bisector:

$$AD = \frac{2bc}{b+c}\cos\frac{A}{2}$$

Orthocentre: $HA = 2R\cos A$, $HD = 2R\cos B\cos C$

$$\angle BHC = \pi - A$$

Pedal: sides $a\cos A$, angles $\pi - 2A$, circumradius $\frac{R}{2}$

Pedal area: $2\Delta\cos A\cos B\cos C$

Excentral: angles $\frac{\pi}{2} - \frac{A}{2}$, sides $4R\cos\frac{A}{2}$, circumradius $2R$

Regular polygon and special figures

Side:

$$a = 2R\sin\frac{\pi}{n} = 2r\tan\frac{\pi}{n}$$

Area:

$$\frac{n}{2}R^2\sin\frac{2\pi}{n} = nr^2\tan\frac{\pi}{n}$$

Interior angle: $\frac{(n-2)\pi}{n}$; diagonal $2R\sin\frac{k\pi}{n}$

Equilateral: $R = 2r$, $r : R : r_1 = 1 : 2 : 3$, $\Delta = \frac{\sqrt{3}}{4}a^2$

Cyclic quadrilateral: $\sqrt{(s-a)(s-b)(s-c)(s-d)}$